

Ahmed Ikram | AI & ML Engineering

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PROFILE

3rd-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Builds production AI systems spanning agentic LLM applications, applied NLP pipelines, and ML lifecycle infrastructure - including a SageMaker-deployed anomaly detection engine with 99.8% cross-validated accuracy and a LangGraph-based financial assistant backed by Airflow retraining, MLflow model management, and cloud deployment on AWS. Seeking a graduate AI Engineering, Applied ML, or MLOps role to contribute production-grade model pipelines, LLM system design, and reliable deployment workflows.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

ATM Log Aggregation, Analysis & Diagnostics Platform | *FastAPI, XGBoost, Isolation Forest, MLflow, SageMaker, LangChain, ChromaDB, Kafka, Redis, Docker, AWS* | Mar 2026 - June 2026

- Backend and ML engineering lead on a 7-person Agile team for NCR Atleos; independently built the 3-layer detection engine, Agentic RAG assistant, full MLOps infrastructure, AWS SageMaker production inference endpoint, and a 1,402-test suite - all post-deadline on top of the team delivered foundation
- 3-layer anomaly detection engine runs every 30s ML_Ensemble (XGBoost + Isolation Forest) on 49 hand-engineered features across 7 groups achieving 99.8% $\pm 0.1\%$ StratifiedKFold CV accuracy across 8 classes and 97.3% IF precision via grid search and Youden's J threshold calibration (F1=0.7008 at -0.51009); ZScore detects distribution shift via rolling 20-window Z-scores independently cross-checked against a deployed SageMaker endpoint (laad-xgb-champion, ml.t2.medium) for cloud-side validation without making inference dependent on cloud availability; all runs and 7 artifacts versioned in MLflow (RDS backend + S3 artifact store with champion aliases)
- Agentic RAG diagnostic assistant built over ChromaDB + Ollama with cross-encoder reranking (+5–15% retrieval relevance over bi-encoder cosine), self-consistency scoring (3 parallel LLM samples via ThreadPoolExecutor, 3-gram Jaccard similarity), reflexion self-critique (generate $\rightarrow$ critique $\rightarrow$ regenerate), and 4-signal confidence fusion (retrieval 30% + consistency 25% + verbalized 25% + grounding 20%) with Platt calibration on ECE trigger; parallelising self-consistency cut end-to-end latency from 26–53s to 11–23s; responses served at <100ms from Redis cache
- 1,402 tests (923 pytest backend at 82% coverage across 10 tiers, 394 vitest frontend, 10 Playwright E2E, 75 Terraform assertions) gating every PR - catching 7 critical defects including a JWT privilege escalation vulnerability and silent data loss under 50 concurrent writes; full AWS estate provisioned via Terraform (10 modules, 118 resources) with 3 GitHub Actions pipelines (OIDC auth, zero static credentials)

StockLens FinTech Spending & Investment App | *Python, FastAPI, Airflow, PostgreSQL, Redis, MLflow, LangGraph, Optuna, PyTorch, Rust, Terraform, AWS* | Oct 2025 - Jul 2026

- LangGraph ReAct financial assistant agent exposing 16 tools across 7 categories to AWS Bedrock Converse with SSE streaming responses, and Elasticache/RDS conversation persistence that survives server restarts with full history reconstruction, and LLM-as-judge evaluation with LangSmith tracing
- Confidence-gated 4-stage OCR and NLP extraction pipeline for receipt understanding, combining Tesseract Regex, Bedrock Vision, text-only LLM, and merchant/category resolution with RapidFuzz matching and a 24h Redis cache keyed on receipt hash avoids repeat inference on duplicate scans
- Weekly automated retraining and production ML operations via Airflow on ECS Fargate: 6-year OHL CV ingestion, feature generation, MLflow-tracked training, champion/challenger, promotion gated at >2pp directional accuracy improvement with delivery to EFS, S3, and a PostgreSQL model registry: Evidently AI drift monitoring via PSI/KS/JSD thresholds with alerting to CloudWatch and reports archived to S3
- Trained a global LSTM for 3-class price-direction forecasting across the S&P 500 tickers, achieving 51.6% directional accuracy against a 33% random baseline and a 0.97 simulated Sharpe ratio; accelerated training and live inference with a custom Rust/PyO3 feature engine replacing pandas-based computation
- Terraform IaC (14 modules, 193 resources) on ECS Fargate ARM64/Graviton behind a WAF-protected ALB, with Multi-AZ RDS, ElastiCache Redis, and CPU/RPS-based autoscaling; validated via 1,415 backend tests and 822 frontend assertions at 90% coverage gates, enforced by a 9-parallel-job CI pipeline and a 7-stage OIDC-authenticated CD pipeline

Haggis Species Classification & Predictive Modelling | Python, Scikit-learn, XGBoost, Matplotlib, Seaborn, Pandas, NumPy, Jupyter Notebook | Nov 2025 - Dec 2025

- Engineered and validated 2 novel ratio-based features that emerged as top-3 predictors across all models; prevented data leakage using sklearn Pipelines with ColumnTransformer and RobustScaler throughout
- Trained and compared 7 classifiers including XGBoost, Random Forest, and Logistic Regression achieving ~90% accuracy; tuned hyperparameters via GridSearchCV with 5-fold cross-validation
- Applied K-Means and DBSCAN clustering to validate natural groupings against ground-truth labels; built a Linear Regression model predicting body mass ($R^2=0.756$), validating assumptions via residual diagnostics

DevSync Project Tracker with GitHub Integration | React, Flask, PostgreSQL, Socket.io, GitHub OAuth, AWS, Docker, CI/CD Jan 2025 - Apr 2026

- Deployed production AWS infrastructure: ECS Fargate + ALB + RDS private subnet + S3 + CloudFront + ACM; rolling deploys tagged by commit SHA for immutable rollback; OIDC federation scoped to main branch with no static credentials stored or transmitted anywhere in the pipeline, repository or task definitions
- 1,452 automated tests at 85%/85% line coverage hard-gate every PR - 7 path-aware CI jobs (lint: ruff + ESLint, security: pip-audit + npm audit, Postgres 15 integration, Cypress E2E) must all pass before deployment. Docker layer caching. Dependabot + CodeQL on weekly cadence

TECHNICAL SKILLS

Languages	Python · SQL · TypeScript · JavaScript · Bash · Rust
ML & AI	PyTorch · Scikit-learn · XGBoost · Isolation Forest · Optuna · Pandas · NumPy
Infra & MLOps	MLflow · Apache Airflow · Terraform · Docker · GitHub Actions · Evidently AI
AI Agents, NLP & RAG	LangGraph · LangChain · ChromaDB · Ollama · Tesseract · RapidFuzz · OCR
Data & Backend	Apache Kafka · FastAPI · Flask · PostgreSQL · Redis · MySQL · MongoDB
Cloud & Tools	AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, SageMaker, Bedrock, CloudFront, CloudWatch, Secrets Manager) · Linux · PyO3 · Maturin
Visualisation	Matplotlib · Seaborn · Jupyter Notebook · Power BI
Testing & Evaluation	Pytest · Vitest · Playwright · dbt Tests · LangSmith

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	Feb 2025
<u>AWS Academy Graduate - Cloud Foundations</u>	Mar 2024
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	Oct 2024

EXPERIENCE

Beaully Mini Market , Dundee, Scotland	
Retail Assistant	September 2023 - Present
• Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption	
Zulus Peri Peri Grill , Dundee, Scotland	
Front of House Worker	June 2021 - September 2023
• Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure	